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CUT-OFF SAW

Abstract

A cut-off saw includes a housing; an electric motor having a motor shaft; an output shaft for mounting and driving a saw blade; a transmission assembly for transmitting power between the motor shaft and the output shaft; a circuit board assembly including a circuit board and a heat dissipation member, where the circuit board is connected to the heat dissipation member; and a heat conductive member connected to the housing, where the heat conductive member is at least partially located outside the housing. The housing has an opening, and the heat dissipation member is at least partially exposed relative to the housing through the opening and is in thermal contact with the heat conductive member.

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Background/Summary

RELATED APPLICATION INFORMATION

[0001] This application claims the benefit under 35 U.S.C. § 119(a) of Chinese Patent Application No. 202410226917.7, filed on Feb. 29, 2024, Chinese Patent Application No. 202411918809.2, filed on Dec. 24, 2024, and Chinese Patent Application No. 202423197085.5, filed on Dec. 24, 2024, which applications are incorporated herein by reference in their entireties.

BACKGROUND

[0002] A cut-off saw in the related art may be handheld and used for cutting concrete or the like. An air-cooling heat dissipation solution is adopted in which an electric motor drives a fan to rotate so that the airflow outside the housing flows through a circuit board, the electric motor, and other heat-generating components inside the housing and then is discharged from the housing. In heavy load conditions, the existing air-cooling solution cannot satisfy the heat dissipation requirements.

[0003] This part provides background information related to the present application, and the background information is not necessarily the existing art.

SUMMARY

[0004] A cut-off saw includes a housing; an electric motor having a motor shaft; an output shaft for mounting and driving a saw blade; a transmission assembly for transmitting power between the motor shaft and the output shaft; a circuit board assembly including a circuit board and a heat dissipation member, where the circuit board is connected to the heat dissipation member; and a heat conductive member connected to the housing, where the heat conductive member is at least partially located outside the housing. The housing has an opening, and the heat dissipation member is at least partially exposed relative to the housing through the opening and is in thermal contact with the heat conductive member.

[0005] In some examples, the heat dissipation member is a heat dissipation box, a side of the heat dissipation box facing away from the circuit board has multiple heat dissipation through holes, and the multiple heat dissipation through holes penetrate a box body of the heat dissipation box along a direction basically parallel to a plane where the circuit board is located.

[0006] In some examples, the opening is located on the front bottom surface of the housing, the heat conductive member is connected to the housing around the opening and has a heat conductive plane covering the opening, and the heat dissipation member is in contact with the heat conductive member through the heat conductive plane.

[0007] In some examples, the cut-off saw further includes a liquid cooling system, where the liquid cooling system sprays coolant onto the saw blade when the saw blade is cutting, and the coolant moves through the heat conductive member after leaving the saw blade.

[0008] In some examples, the liquid cooling system includes a transfer pipe for transmitting coolant, a groove for the transfer pipe to pass through is formed inside the housing, and an extension path of the transfer pipe passes through the housing around the periphery of the circuit board assembly.

[0009] In some examples, the projection of the transfer pipe and the projection of the circuit board in the up and down direction at least partially coincide; and/or the projection of the transfer pipe and the projection of the heat dissipation member in the up and down direction at least partially coincide.

[0010] In some examples, the cut-off saw further includes a sealing member, where the sealing member is disposed around the opening and fills the gaps between the housing, the heat dissipation member, and the heat conductive member.

[0011] In some examples, the cut-off saw further includes a fan, where the fan is mounted on the motor shaft, the housing has an air inlet and an air outlet, and the cooling airflow enters the housing from the air inlet, flows through the circuit board assembly and the electric motor in sequence, and is discharged from the air outlet.

[0012] In some examples, the air inlet is located on the left wall and/or the right wall of the middle and rear part of the housing, and the air outlet is located at the front part of the housing.

[0013] In some examples, the heat conductive member is a wear-resistant metal member.

[0014] A cut-off saw includes a housing; an electric motor having a motor shaft; an output shaft for mounting and driving a saw blade; and a transmission assembly for transmitting power between the motor shaft and the output shaft. The cut-off saw further includes a circuit board assembly including a circuit board and a heat dissipation member, where the circuit board is connected to the heat dissipation member. The housing has an opening, and the heat dissipation member is at least partially exposed relative to the housing through the opening.

[0015] In some examples, the housing includes a body portion extending along the front and rear direction, the body portion has a support surface for supporting the cut-off saw when the cut-off saw is placed horizontally, the motor shaft extends along a motor axis, and the motor axis obliquely intersects with the support surface.

[0016] In some examples, the output shaft is located outside the housing, and the output shaft is basically parallel to the support surface and extends along the left and right direction of the housing.

[0017] In some examples, the cut-off saw further includes a second housing, where the second housing is connected to the housing and is located outside the housing, the transmission assembly is at least partially located in the second housing, and the central axis of the second housing obliquely intersects with a support surface.

[0018] In some examples, the transmission assembly includes a long shaft and a bevel gearset, the long shaft is connected to and collinear with the motor shaft, and the bevel gearset is connected between the long shaft and the output shaft to transmit power.

[0019] In some examples, the cut-off saw further includes a shaft locking assembly disposed on the second housing, and the shaft locking assembly includes an operating member and a shaft locking member connected to each other. Shaft locking grooves are circumferentially disposed on the long shaft. The shaft locking member may be embedded in the shaft locking groove. The shaft locking grooves extend along a direction basically parallel to the long shaft. The operating member is operated by the user to push the operating member and the shaft locking member along the direction basically parallel to the long shaft so that the shaft locking member is embedded in the shaft locking groove, thereby locking the rotation of the output shaft.

[0020] In some examples, the cut-off saw further includes a heat conductive member, where the heat conductive member, the heat dissipation member, and the second housing are connected; the heat dissipation member, the heat conductive member, the second housing, and the saw blade are all made of conductive materials, and the static electricity generated in a cutting process is transferred to the saw blade through the heat dissipation member, the heat conductive member, and the second housing and then released to the ground.

[0021] In some examples, the cut-off saw further includes a guard and a guard adjustment mechanism, and the guard is connected to the second housing through the guard adjustment mechanism. The guard adjustment mechanism includes an adjustment bolt and an adjustment handle. A rotation track is disposed on a side of the guard facing the second housing. An end of the adjustment bolt is fixedly connected to the second housing, and the other end of the adjustment bolt is rotatably connected to the guard through the rotation track. The adjustment handle releases or locks the relative rotation between the adjustment bolt and the guard, thereby locking or releasing the relative rotation between the second housing and the guard.

[0022] In some examples, the guard adjustment mechanism performs stepless adjustment on the

angle of rotation of the guard about the output shaft. The guard is rotatable about the output shaft within a range of 0° to 45°.

[0023] In some examples, the circuit board is located below the electric motor, and the projection of the circuit board and the projection of the electric motor in the up and down direction at least partially coincide.

[0024] A cut-off saw includes a housing; an electric motor having a motor shaft; an output shaft for mounting and driving a saw blade; and a transmission assembly for transmitting power between the motor shaft and the output shaft. The cut-off saw further includes a circuit board assembly including a circuit board and a heat dissipation box, where the circuit board is connected to the heat dissipation box, a side of the heat dissipation box facing away from the circuit board has multiple heat dissipation through holes, and the multiple heat dissipation through holes penetrate a box body of the heat dissipation box along a direction basically parallel to a plane where the circuit board is located; and an air guide plate connected to the housing and located inside the housing. The housing has an air inlet, where the housing, the air guide plate, and the heat dissipation box form a first airflow channel between the air inlet and the multiple heat dissipation through holes, and the cooling airflow enters the housing from the air inlet and passes through the first airflow channel and the multiple heat dissipation through holes in sequence.

[0025] In some examples, the housing has the air outlet, and the electric motor has a motor housing. The housing and the motor housing form a second airflow channel between the electric motor and the air outlet, and the cooling airflow flows through the electric motor in the motor housing and is discharged from the air outlet through the second airflow channel.

[0026] In some examples, the first airflow channel is basically sealed between the air inlet and the multiple heat dissipation through holes, and/or the second airflow channel is basically sealed between the electric motor and the air outlet.

[0027] In some examples, the heat dissipation box is located below the electric motor, the projection of the heat dissipation box and the projection of the electric motor in the up and down direction at least partially coincide, and the cooling airflow passes through the heat dissipation through holes and then enters the motor housing through the inner cavity of the housing.

[0028] In some examples, the cut-off saw further includes a guard and a liquid cooling system, the saw blade is partially accommodated in the guard, and the liquid cooling system sprays coolant onto the saw blade when the saw blade is cutting. The liquid cooling system includes a nozzle, and the nozzle penetrate the guard to output the coolant.

[0029] In some examples, the nozzle includes a left nozzle and a right nozzle, and the left nozzle is disposed on the left end surface of the guard, the right nozzle is disposed on the right end surface of the guard.

[0030] In some examples, the liquid cooling system further includes a liquid inlet valve and a transfer pipe, where the transfer pipe is used for transferring the coolant, and the liquid inlet valve adjusts the flow rate of the coolant in the transfer pipe.

[0031] In some examples, the left nozzle and/or the right nozzle each include a first member and a second member, the first member is connected to the guard, a through hole penetrating the guard exists, and the second member is detachably connected to the first member. In a state in which the second member is mounted to the first member, an end of the through hole facing away from the guard is closed by the second member. In a state in which the second member is disassembled from the first member, the end of the through hole facing away from the guard is opened.

[0032] In some examples, the cut-off saw further includes a second housing located outside the housing, and the transmission assembly is at least partially accommodated in the second housing; the liquid cooling system further includes a transfer pipe for transferring coolant, and the transfer pipe extends from the front end of the housing, passes through the gap between the second housing and the guard, crosses over the guard, and is connected to the left nozzle and the right nozzle.

[0033] In some examples, the upper part of the guard is provided with a guard handle for the user

to operate to rotate the guard. When the transfer pipe crosses over the guard, the transfer pipe is limited by the guard handle.

[0034] In some examples, the cut-off saw further includes a support guide member. The support guide member has a support portion located between the second housing and the guard. The transfer pipe passes through the support portion and is limited by the support portion.

[0035] A cut-off saw includes a housing; an electric motor having a motor shaft; an output shaft for mounting and driving a saw blade; and a transmission assembly for transmitting power between the motor shaft and the output shaft. The cut-off saw further includes a battery pack. The battery pack supplies power to at least the electric motor, the housing includes a body portion and a battery compartment cover, a battery compartment is formed inside the body portion, and the battery compartment cover has a closed state and an open state. In the closed state, the battery compartment cover seals the battery compartment; and in the open state, the battery pack is detachably inserted into or removed from the battery compartment along the left and right direction.

[0036] In some examples, the battery compartment cover is rotatably connected to the body portion. The battery compartment cover is rotatable about a pivot axis, and the pivot axis is perpendicular to the bottom surface of the housing.

[0037] In some examples, the battery compartment cover in the open state can freely rotate about the pivot axis with an angle of 0° to 180° relative to the closed state.

[0038] In some examples, the battery compartment cover in the open state rotates about the pivot axis to a preset angle relative to the closed state and remains at the preset angle, where the preset angle is greater than or equal to 90° and less than or equal to 180° .

[0039] In some examples, one of the body portion and the battery compartment cover is provided with a second snap catch, the other one of the body portion and the battery compartment cover is provided with a second snap groove, and the second snap catch mates with the second snap groove, thereby sealing the battery compartment. The battery compartment cover is rotatably connected to the body portion via a connecting shaft. The connecting shaft extends along the pivot axis. The connecting shaft is connected to an elastic member. When the second snap catch disengages from the second snap groove, the elastic member causes the battery compartment cover to rotate about the pivot axis to the preset angle relative to the closed state and remain at the preset angle.

[0040] In some examples, one of the battery pack and the battery compartment is provided with a first snap catch, and the other one of the battery pack and the battery compartment is provided with a first snap groove. The first snap catch mates with the first snap groove so that the battery pack can be locked to the battery compartment. The cut-off saw further includes a battery pack ejection mechanism disposed on the inner wall of the battery compartment. When the first snap catch disengages from the first snap groove, at least part of the battery pack pops up from the battery compartment through the battery pack ejection mechanism along the insertion and removal direction.

[0041] In some examples, a circuit wire and a terminal block are provided on the inner wall of the battery compartment. The cut-off saw further includes a circuit cover plate disposed in the battery compartment. The circuit cover plate mates with the inner wall of the battery compartment, thereby limiting the circuit wire.

[0042] In some examples, the terminal block is disposed on the upper inner wall of the battery compartment; the circuit wire and the circuit cover plate are disposed on the left inner wall or the right inner wall of the battery compartment facing away from the battery compartment cover.

[0043] A cut-off saw includes a housing; an electric motor having a motor shaft; an output shaft for mounting and driving a saw blade; a transmission assembly for transmitting power between the motor shaft and the output shaft; and a battery pack for supplying power to at least the electric motor. The housing of the battery pack is provided with an indicator for displaying the state information of the battery pack. A battery compartment is formed inside the housing. The battery

pack is detachably inserted into or removed from the battery compartment. The housing has a transparent portion, and the transparent portion is aligned with the indicator so that the state information displayed by the indicator can be observed through the transparent portion.

[0044] In some examples, the indicator displays at least the remaining power information of the battery pack.

[0045] In some examples, the indicator is the remaining power indicator light for the battery pack.

[0046] In some examples, the housing includes a body portion, a grip, and a battery compartment cover. The grip is at least partially located behind the body portion. The battery compartment cover seals the battery compartment. The transparent portion is located on the upper rear side of the body portion and in front of the grip, and/or the transparent portion is located on the battery compartment cover.

[0047] In some examples, the housing is provided with a first operating member, and the first operating member is operated by the user to make the indicator display the state information.

[0048] In some examples, the housing of the battery pack is further provided with a second operating member for the user to operate so that the indicator displays the state information. The second operating member is connected to the first operating member.

[0049] In some examples, the housing is provided with a battery indicator, and the battery indicator is electrically connected to the circuit board to acquire and display the state information of the battery pack.

[0050] In some examples, the battery indicator includes an operating portion, an indicator portion, and a sub-circuit board. The sub-circuit board is electrically connected to the operating portion, the indicator portion, and the circuit board. In response to the signal sent by the operating portion operated by the user, the sub-circuit board acquires the state information of the battery pack from the circuit board and triggers the indicator portion to display the corresponding information.

[0051] In some examples, the operating portion and the indicator portion may be embedded in the surface of the housing, and the sub-circuit board may be disposed on the back side of the operating portion and the indicator portion and located inside the housing.

[0052] In some examples, the sub-circuit board and the circuit board may be electrically connected by a flat cable, and the flat cable may be attached to the inner wall of the housing.

[0053] In some examples, the state information of the battery pack includes the remaining power information and/or fault warning information of the battery pack.

[0054] In some examples, the indicator portion includes a remaining power indicator light or a display screen.

[0055] A cut-off saw includes a housing; an electric motor having a motor shaft; an output shaft for mounting and driving a saw blade; a transmission assembly for transmitting power between the motor shaft and the output shaft; and a battery pack for supplying power to at least the electric motor. The housing is provided with a battery indicator, and the battery indicator is electrically connected to the circuit board to acquire and display the state information of the battery pack.

[0056] A cut-off saw includes a housing; an electric motor having a motor shaft; an output shaft for mounting and driving a saw blade; a transmission assembly for transmitting power between the motor shaft and the output shaft; and a second housing located outside the housing and accommodating at least part of the transmission assembly. The second housing is provided with a through hole to allow the airflow and/or lubricant to flow through. The cut-off saw further includes an expansion cover. The expansion cover covers the through hole and is sealingly connected to the second housing, thereby forming a sealed space with variable volume that is opened only to the through hole and is between the expansion cover and the second housing.

[0057] In some examples, the expansion cover is made of a soft rubber material, and the soft rubber material includes rubber or silicone.

[0058] In some examples, the second housing includes a second housing body and a first end cover, the first end cover and the second housing body are sealing connected to form a first

accommodation space for accommodating at least the transmission assembly, the first end cover is provided with a through hole, and the expansion cover is mounted to the first end cover.

[0059] In some examples, the second housing further includes a second end cover, the second end cover is mounted to the first end cover, and the expansion cover is sandwiched between the first end cover and the second end cover.

[0060] In some examples, a second accommodation space for accommodating at least a bearing assembly is formed between the second end cover and the first end cover.

[0061] In some examples, a docking portion of the first end cover and a docking portion of the second end cover are opposite to each other, and the expansion cover is clamped and fixed between the docking portions.

[0062] In some examples, the docking portions partition the second accommodation space into a space where the bearing assembly is located and a space where the expansion cover is located.

[0063] In some examples, the second end cover is provided with an opening connecting with the outside, and the expansion cover divides the second accommodation space between the second end cover and the first end cover into at least a sealed space connecting with the first accommodation space through the through hole on the first end cover and an open space connecting with the outside through the opening on the second end cover.

[0064] In some examples, in the case of an increase in temperature and/or pressure within the first accommodation space that is between the first end cover and the second housing body and accommodates the transmission assembly, the gas and/or lubricant in the first accommodation space enters the gap between the expansion cover and the first end cover via the through hole.

[0065] In some examples, an end of the output shaft and the transmission assembly are connected in the first accommodation space, and after the output shaft passes through the first end cover and the second end cover, the other end of the output shaft enters the guard and is used for mounting and driving the saw blade.

[0066] A cut-off saw includes a housing; an electric motor having a motor shaft; an output shaft for mounting and driving a saw blade; and a transmission assembly for transmitting power between the motor shaft and the output shaft. The cut-off saw further includes a hook assembly through which the entire cut-off saw is suspended in a basically vertical posture at a support point.

[0067] In some examples, the hook assembly is disposed on the bottom surface of the housing.

[0068] In some examples, the hook assembly is disposed at a higher portion of the bottom surface of the housing along the up and down direction.

[0069] In some examples, the housing includes a body portion extending along the front and rear direction, a battery compartment is formed inside the body portion, and the hook assembly is disposed on the bottom surface of the housing below the battery compartment.

[0070] In some examples, the housing includes a main handle located at the rear of the cut-off saw, and the hook assembly is disposed on the bottom surface of the main handle or at the rear end of the main handle.

[0071] In some examples, the hook assembly has a storage state and a suspended state, the hook assembly is rotatably connected to the housing, and the hook assembly rotates relative to the housing to switch between the storage state and the suspended state.

[0072] In some examples, the hook assembly includes a hook body and a mounting member, the mounting member is fixedly connected to the housing, and the hook body is rotatably connected to the mounting member.

[0073] In some examples, the hook body in the suspended state may be rotated about the mounting member by at least 90° relative to the hook body in the storage state.

[0074] In some examples, the bottom surface of the housing of the cut-off saw in the suspended state is basically perpendicular to the ground, and the surface where the hook body is located is basically perpendicular to the bottom surface of the housing, where the bottom surface of the housing when the cut-off saw is placed horizontally is basically parallel to the ground.

[0075] In some examples, in the suspended state, the surface where the hook body is located is basically parallel to the bottom surface of the housing.

[0076] A cut-off saw includes a housing; an electric motor having a motor shaft; an output shaft for mounting and driving a saw blade; and a transmission assembly for transmitting power between the motor shaft and the output shaft. The cut-off saw further includes a shaft locking assembly having a locking state for locking the rotation of the output shaft and an unlocking state for releasing the rotation of the output shaft and including a shaft locking member, an elastic member, and a support limiting member. The transmission assembly has shaft locking grooves. The shaft locking member can be embedded in the shaft locking groove to block the power transmission to the output shaft. The elastic member abuts against the shaft locking member, and the shaft locking member always tends to be embedded in the shaft locking groove due to the elastic deformation of the elastic member. The shaft locking member is configured to rotate upward or downward relative to the support limiting member under the constraint of the support limiting member, thereby achieving the switching of the shaft locking assembly between the locking state and the unlocking state.

[0077] In some examples, the shaft locking assembly further includes an operating member connected to or formed on the shaft locking member, and the operating member rotates to switch between a locking position and an unlocking position.

[0078] In some examples, the rotation axis of the operating member is collinear with or parallel to the rotation axis of the shaft locking member.

[0079] In some examples, the support limiting member has a helical track along the circumferential direction of the shaft locking member, the shaft locking member is connected to or formed with a mating portion, and under the constraint of the track, the mating portion moves along the helical track to drive the shaft locking member to rotate upward or downward.

[0080] In some examples, the helical track of the support limiting member is at least partially accommodated in the operating member.

[0081] In some examples, locking portions are provided at two ends of the helical track. When the mating portion moves to one of the two ends of the helical track, the mating portion is limited by the locking portion, thereby maintaining the shaft locking assembly in the locking state or the unlocking state.

[0082] In some examples, when the operating member is rotated to switch from one of the locking position and the unlocking position to the other one of the locking position and the unlocking position, the mating portion overcomes the limit of the locking portion at an end of the helical track and is driven by the elastic member or overcomes the drive of the elastic member to rotate upward or downward along the helical track until the mating portion reaches the locking portion at the other end of the helical track and limited and stopped by the locking portion. In this manner, the shaft locking member can disengage from or be embedded in the shaft locking groove.

[0083] In some examples, the cut-off saw further includes a second housing located outside the housing and accommodating at least part of the transmission assembly, and the support limiting member is fixedly mounted on the second housing.

[0084] In some examples, the transmission assembly includes a long shaft connected to the motor shaft, and shaft locking grooves are disposed along the circumferential direction of the long shaft.

[0085] In some examples, a guide surface is provided between adjacent shaft locking grooves; the shaft locking member rotates upward or downward to the guide surface, slides relative to the shaft locking groove along the guide surface, and is embedded in the shaft locking groove.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0086] FIG. 1 is a perspective view of a cut-off saw according to the present application from one

perspective.

[0087] FIG. 2 is a perspective view of a cut-off saw according to the present application from another perspective.

[0088] FIG. 3 is a perspective view of a structure in which a heat conductive member and a heat dissipation member in the cut-off saw shown in FIG. 2 are separated.

[0089] FIG. 4 is an exploded view of a cut-off saw according to the present application.

[0090] FIG. 5 is a perspective view of an electric motor, a transmission assembly, an output shaft, and a shaft locking assembly in a cut-off saw according to the present application.

[0091] FIG. 6 is a plan view of the shaft locking assembly in the cut-off saw shown in FIG. 5 in an unlocking state.

[0092] FIG. 7 is a plan view of the shaft locking assembly in the cut-off saw shown in FIG. 5 in a locking state.

[0093] FIG. 8 is a left side view of a cut-off saw according to the present application.

[0094] FIG. 9 is a perspective view of a cut-off saw with part of a housing removed according to the present application from one perspective.

[0095] FIG. 10 is a perspective view of a cut-off saw with part of a housing removed according to the present application from another perspective.

[0096] FIG. 11 is schematic view one illustrating the cooling airflow direction in a cut-off saw according to the present application.

[0097] FIG. 12 is schematic view two illustrating the cooling airflow direction in a cut-off saw according to the present application.

[0098] FIG. 13 is a perspective view of a structure in which part of a housing, an air guide plate, and a heat dissipation member in the cut-off saw shown in FIG. 12 are separated.

[0099] FIG. 14 is schematic view three illustrating the cooling airflow direction in a cut-off saw according to the present application.

[0100] FIG. 15 is a right side view of a cut-off saw with part of a housing removed according to the present application.

[0101] FIG. 16 is a sectional view of the cut-off saw shown in FIG. 15 taken along A-A.

[0102] FIG. 17 is a perspective view of an electric motor, a second housing, a guard, and a guard adjustment mechanism in a cut-off saw according to the present application.

[0103] FIG. 18 is an exploded view of the guard and the guard adjustment mechanism in the cut-off saw shown in FIG. 17.

[0104] FIG. 19 is a perspective view of a structure in which part of a housing, a heat dissipation member, a sealing member, and a heat conductive member in a cut-off saw are separated according to the present application.

[0105] FIG. 20 is an exploded view of a left nozzle in a cut-off saw according to the present application.

[0106] FIG. 21 is a plan view illustrating that a guard in a cut-off saw is at a first position according to the present application.

[0107] FIG. 22 is a plan view illustrating that a guard in a cut-off saw is at a second position according to the present application.

[0108] FIG. 23 is a perspective view of a battery compartment, a battery compartment cover, and a battery pack in a cut-off saw according to the present application.

[0109] FIG. 24 is a perspective view of a structure in which the battery compartment and the battery compartment cover in the cut-off saw shown in FIG. 23 are separated.

[0110] FIG. 25 is an exploded view of some structures in the battery compartment in the cut-off saw shown in FIG. 23.

[0111] FIG. 26 is a perspective view of a cut-off saw according to the present application from another perspective.

[0112] FIG. 27 is a perspective view of a battery indicator and a circuit board assembly of the cut-

off saw shown in FIG. 26 with part of a housing removed.

[0113] FIG. 28 is a perspective view of the exploded structure of the battery indicator shown in FIG. 27.

[0114] FIG. 29 is a perspective view of a transfer pipe and an elastic support member in the cut-off saw shown in FIG. 26.

[0115] FIG. 30 is a perspective view of a structure in which a left nozzle, a guard, and a right nozzle in the cut-off saw shown in FIG. 26 are separated.

[0116] FIG. 31 is a perspective view of a structure in which a second housing body, a first end cover, an expansion cover, a second end cover, and a guard in the cut-off saw shown in FIG. 26 are separated.

[0117] FIG. 32 is a sectional view of a guard, an output shaft, a bearing assembly, and a second housing of the cut-off saw shown in FIG. 26.

[0118] FIG. 33 is a plan view of a hook assembly in the cut-off saw shown in FIG. 26 in a storage state.

[0119] FIG. 34 is a plan view of a hook assembly in the cut-off saw shown in FIG. 26 in a suspended state.

[0120] FIG. 35 is a perspective view of a shaft locking assembly in a locking state with part of a housing of the cut-off saw shown in FIG. 26 removed.

[0121] FIG. 36 is a perspective view of a shaft locking assembly in an unlocking state with part of a housing of the cut-off saw shown in FIG. 26 removed.

[0122] FIG. 37 is an exploded view of the shaft locking assembly shown in FIG. 35.

[0123] FIG. 38 is a sectional view of the shaft locking assembly shown in FIG. 35.

DETAILED DESCRIPTION

[0124] Before any examples of this application are explained in detail, it is to be understood that this application is not limited to its application to the structural details and the arrangement of components set forth in the following description or illustrated in the above drawings.

[0125] In this application, the terms “comprising”, “including”, “having” or any other variation thereof are intended to cover an inclusive inclusion such that a process, method, article or device comprising a series of elements includes not only those series of elements, but also other elements not expressly listed, or elements inherent in the process, method, article, or device. Without further limitations, an element defined by the phrase “comprising a . . .” does not preclude the presence of additional identical elements in the process, method, article, or device comprising that element.

[0126] In this application, the term “and/or” is a kind of association relationship describing the relationship between associated objects, which means that there can be three kinds of relationships. For example, A and/or B can indicate that A exists alone, A and B exist simultaneously, and B exists alone. In addition, the character “/” in this application generally indicates that the contextual associated objects belong to an “and/or” relationship.

[0127] In this application, the terms “connection”, “combination”, “coupling” and “installation” may be direct connection, combination, coupling or installation, and may also be indirect connection, combination, coupling or installation. Among them, for example, direct connection means that two members or assemblies are connected together without intermediaries, and indirect connection means that two members or assemblies are respectively connected with at least one intermediate members and the two members or assemblies are connected by the at least one intermediate members. In addition, “connection” and “coupling” are not limited to physical or mechanical connections or couplings, and may include electrical connections or couplings.

[0128] In this application, it is to be understood by those skilled in the art that a relative term (such as “about”, “approximately”, and “substantially”) used in conjunction with quantity or condition includes a stated value and has a meaning dictated by the context. For example, the relative term includes at least a degree of error associated with the measurement of a particular value, a tolerance caused by manufacturing, assembly, and use associated with the particular value, and the like. Such

relative term should also be considered as disclosing the range defined by the absolute values of the two endpoints. The relative term may refer to plus or minus of a certain percentage (such as 1%, 5%, 10%, or more) of an indicated value. A value that did not use the relative term should also be disclosed as a particular value with a tolerance. In addition, “substantially” when expressing a relative angular position relationship (for example, substantially parallel, substantially perpendicular), may refer to adding or subtracting a certain degree (such as 1 degree, 5 degrees, 10 degrees or more) to the indicated angle.

[0129] In this application, those skilled in the art will understand that a function performed by an assembly may be performed by one assembly, multiple assemblies, one member, or multiple members. Likewise, a function performed by a member may be performed by one member, an assembly, or a combination of members.

[0130] In this application, the terms “up”, “down”, “left”, “right”, “front”, and “rear” and other directional words are described based on the orientation or positional relationship shown in the drawings, and should not be understood as limitations to the examples of this application. In addition, in this context, it also needs to be understood that when it is mentioned that an element is connected “above” or “under” another element, it can not only be directly connected “above” or “under” the other element, but can also be indirectly connected “above” or “under” the other element through an intermediate element. It should also be understood that orientation words such as upper side, lower side, left side, right side, front side, and rear side do not only represent perfect orientations, but can also be understood as lateral orientations. For example, lower side may include directly below, bottom left, bottom right, front bottom, and rear bottom.

[0131] Referring to FIGS. **1** to **10**, the present application provides a cut-off saw **100**. The cut-off saw **100** includes a housing **110**, an electric motor **120**, an output shaft **130**, and a transmission assembly **140**. The housing **110** includes a body portion **111** extending along the front and rear direction. The electric motor **120** is disposed in the body portion **111**. The electric motor **120** has a motor shaft **121**. The output shaft **130** is used for mounting and driving a saw blade **150**. The transmission assembly **140** transmits power between the motor shaft **121** and the output shaft **130**.

[0132] The housing **110** further includes a grip at least partially located behind the body portion **111**. The grip includes a main handle **112** located behind the body portion **111**, thereby facilitating holding. In some examples, the main handle **112** forms a through hole through which a hand can pass for holding. The grip further includes a front handle **113** disposed on the body portion **111** and located above the body portion **111**. During use, the front handle **113** may be held by one hand, and the main handle **112** may be held by the other hand, thereby facilitating the operation of the cut-off saw **100**.

[0133] In some examples, the housing **110** includes a left housing and a right housing, and the left housing and the right housing snap together to form the body portion **111** and the grip. In some examples, the body portion **111** is detachably connected to the grip, and the body portion **111** includes a left housing and a right housing. Of course, it is also feasible that various parts of the housing **110** may be integrally formed.

[0134] The body portion **111** has a support surface for supporting the cut-off saw **100** when the cut-off saw **100** is placed horizontally so that the cut-off saw **100** is more stable when placed horizontally. The motor shaft **121** extends along a motor axis **122**, and the motor axis **122** obliquely intersects with the support surface, which is compatible with output mechanisms such as the transmission assembly **140** to be described later, thereby effectively reducing the length of the whole machine. In some examples, the included angle between the motor axis **122** and the support surface is 15° to 60°.

[0135] The output shaft **130** is located outside the housing **110**. The output shaft **130** is parallel to the support surface and extends along the left and right direction of the housing **110**. The output shaft **130** extends along an output axis **131**, and the output axis **131** is parallel to the support surface. The output axis **131** may intersect with the motor axis **122**. In some examples, the motor

shaft **121** is basically perpendicular to the output shaft **130**, that is, the motor axis **122** is perpendicular to the output axis **131**. In some examples, an included angle exists between the motor axis **122** and the output axis **131**.

[0136] The transmission assembly **140** includes a long shaft **141** and a bevel gearset **142**. The long shaft **141** is connected to and collinear with the motor shaft **121**. The bevel gearset **142** is connected between the long shaft **141** and the output shaft **130** to transmit power. The electric motor **120** drives the long shaft **141** to rotate and then transmits power to the output shaft **130** through the bevel gearset **142**.

[0137] The bevel gearset **142** includes a first bevel gear **1421** and a second bevel gear **1422**. The first bevel gear **1421** is disposed on the long shaft **141**, the second bevel gear **1422** is disposed on the output shaft **130**, and the first bevel gear **1421** meshes with the second bevel gear **1422**. In some examples, the first bevel gear **1421** and the long shaft **141** are integrally formed.

[0138] The cut-off saw **100** further includes a second housing **160**, the second housing **160** is connected to the housing **110** and is at least partially located outside the housing **110**, the second housing **160** may be located between the housing **110** and the output shaft **130** or between the housing **110** and the saw blade **150**, and the transmission assembly **140** is at least partially located in the second housing **160**. In some examples, the second housing **160** for accommodating the transmission assembly **140** forms a gearbox and a bearing chamber. The second housing **160** is provided to facilitate assembly, replacement, and maintenance. The central axis of the second housing **160** obliquely intersects with the support surface. The second housing **160** extends along the central axis, and the central axis is parallel to the axis of the long shaft **141**. In some examples, the central axis coincides with the axis of the long shaft **141**.

[0139] The second housing **160** or the gearbox for accommodating the transmission assembly **140** is filled with semi-solid grease or the like for lubrication and is usually a sealed space. However, due to the influence of working conditions, such as high-power operation or high ambient temperature, the temperature and pressure in the cavity increase, and the lubricant is prone to leakage at weak sealing points. To address the preceding problem, in some examples, the second housing **160** is provided with a through hole **165** to allow the airflow and/or lubricant to flow through, and the second housing **160** is further provided with an expansion cover **164** that can cover the through hole **165**. The expansion cover **164** is sealingly connected to the second housing **160**, and the expansion cover **164** is elastically deformable, thereby forming a sealed space with variable volume that is impermeable to both air and oil outside the through hole **165** of the second housing **160**. Therefore, in the case of an increase in temperature and/or air pressure, the gas and/or lubricant in the second housing **160** or the gearbox can enter the sealed space with variable volume formed by the expansion cover **164** and the second housing **160** from the through hole **165** so that the air pressure within the cavity is balanced with the external atmospheric pressure, and the lubricant does not leak, thereby ensuring that the cut-off saw **100** can always work smoothly. In some examples, the expansion cover **164** may be made of a soft rubber material such as rubber or silicone or may be made of other soft and elastic materials that are impermeable to both air and oil.

[0140] In some examples, referring to FIGS. **31** and **32**, the second housing **160** includes a second housing body **161**, a first end cover **162**, and a second end cover **163**. The first end cover **162** may be mounted to and sealingly connected to the second housing body **161**, and a first accommodation space is formed between the first end cover **162** and the second housing body **161**, that is, an inner cavity of the gearbox for accommodating the transmission assembly **140** is formed, and the long shaft **141** and the bevel gearset **142** may be disposed in the gearbox inner cavity. The through hole **165** may be provided on the first end cover **162**, and the expansion cover **164** may cover the through hole **165** and be sealingly connected to the first end cover **162**. The second end cover **163** may be mounted to the first end cover **162**, and a second accommodation space is formed between the second end cover **163** and the first end cover **162**. In some examples, the second accommodation space may accommodate at least a bearing assembly for supporting the output

shaft **130**, that is, an inner cavity of the bearing chamber for accommodating the bearing assembly of the output shaft **130** is formed.

[0141] The first end cover **162** and the second end cover **163** are located between the second housing body **161** and a guard **220**, and the second end cover **163** is closer to the guard **220**. The first end cover **162** and the second end cover **163** are each provided with a through hole for the output shaft **130** to pass through. The output shaft **130** may be connected to the second bevel gear **1422** in the inner cavity of the gearbox, pass through the inner cavity of the gearbox and the inner cavity of the bearing chamber in sequence via the through holes on the first end cover **162** and the second end cover **163**, and then enter the guard **220** so that the saw blade **150** can be mounted on the output shaft **130**, and the output shaft **130** can drive the saw blade **150**.

[0142] The expansion cover **164** may be disposed in the second accommodation space formed between the first end cover **162** and the second end cover **163**. In some examples, as shown in FIG. **32**, a docking portion **166** of the first end cover **162** and a docking portion **166** of the second end cover **163** are opposite to each other and used for clamping and fixing the expansion cover **164**. When the second end cover **163** is mounted on the first end cover **162**, the docking portions **166** clamp the expansion cover **164** therebetween. Moreover, the docking portions **166** further divide the second accommodation space formed between the first end cover **162** and the second end cover **163** into a space that accommodates the bearing assembly and does not need to connect with the outside or the gearbox and a space for accommodating the expansion cover **164** and connecting with the outside and the gearbox. The space for accommodating the expansion cover **164** and connecting with the outside and the gearbox, that is, the space where the expansion cover **164** is located, is partitioned by the expansion cover **164**, a side of the expansion cover **164** and the first end cover **162** form a sealed space that connects with the gearbox and is opened only to the through hole **165** on the first end cover **162** through which the air/oil flows, and the other side of the expansion cover **164**, the second end cover **163**, and an opening **167** on the second end cover **163** form an open space connecting with the outside.

[0143] Referring to FIGS. **4** to **7**, the cut-off saw **100** further includes a shaft locking assembly **170** disposed on the second housing **160**, and the shaft locking assembly **170** includes an operating member **171** and a shaft locking member **172** connected to each other. Shaft locking grooves **1411** are circumferentially disposed on the long shaft **141**. The shaft locking member **172** may be embedded in the shaft locking groove **1411**. The shaft locking grooves **1411** extend along a direction parallel to the long shaft **141**. In some examples, the shaft locking grooves **1411** are disposed on a component sleeved on the long shaft **141**, and the operating member **171** is operated by the user to push the operating member **171** and the shaft locking member **172** along the direction parallel to the long shaft **141** so that the shaft locking member **172** is embedded in the shaft locking groove **1411**, thereby locking the rotation of the output shaft **130**.

[0144] The shaft locking assembly **170** has an unlocking state and a locking state. In the locking state, the output shaft **130** may be locked by the shaft locking assembly **170** so that the transmission assembly **140** does not rotate with the saw blade, thereby facilitating the installation and removal of the saw blade; and the saw blade **150** not in use can be prevented from rotating due to inadvertent contact, thereby ensuring safety. The operating member **171** is of a push type. The operating member **171** can cause the shaft locking member **172** to be embedded in the shaft locking groove **1411** so that the rotation of the long shaft **141** can be locked, thereby further locking the rotation of the output shaft **130**. The direction along which the operating member **171** is pushed is parallel to the long shaft **141**. When the cut-off saw **100** is used, the shaft locking assembly **170** can be switched to the unlocking state by pushing the operating member **171**, which is easy to operate.

[0145] The operating member **171** at least partially protrudes from the second housing **160** so that the user can operate the shaft locking assembly **170** easily. The second housing **160** is provided with an avoidance opening for the operating member **171** to extend out. The operating member **171** is elastically connected to the second housing **160**. The operating member **171** is slidable along a

direction parallel to the long shaft **141**. The avoidance opening extends along the direction of the long shaft **141**. The operating member **171** is provided with a protrusion that protrudes out of the housing **110** through the avoidance opening. The protrusion is provided so that the user can easily push the operating member **171**.

[0146] Referring to FIGS. **35** to **38**, another alternative example of the shaft locking assembly **170** is provided. As shown in FIG. **35**, similar to the above, the shaft locking assembly **170** includes the shaft locking member **172** that can be embedded in the shaft locking groove **1411** to lock the rotation of the output shaft **130**. For example, the shaft locking grooves **1411** may be connected to or formed on the long shaft **141** of the transmission assembly **140**. Specifically, a shaft locking disc on which the shaft locking grooves **1411** are provided may be sleeved on the long shaft **141**. Of course, in other examples, the shaft locking grooves **1411** may be directly disposed on the motor shaft **121** or the output shaft **130**. The shaft locking assembly **170** further includes a support limiting member **173** and an elastic member **174**. The elastic member **174** abuts against the shaft locking member **172**, and the shaft locking member **172** always tends to be embedded in the shaft locking groove **1411** due to the elastic deformation of the elastic member **174**. For example, the elastic member **174** may be a spring that is always in a compressed state or a stretched state, and an end of the elastic member **174** is fixed. The shaft locking member **172** rotates upward or downward under the constraint of the support limiting member **173** and the drive of the elastic member **174** to disengage from or be embedded in the shaft locking groove **1411**, thereby achieving the switching of the shaft locking assembly **170** between the unlocking state and the locking state.

[0147] In this example, the operating member **171** may be of a rotary type and may be connected to or formed on the shaft locking member **172**. The operating member **171** rotates to switch between an unlocking position and a locking position, and the shaft locking assembly **170** may convert the rotation of the operating member **171** into the rotational ascent or descent of the shaft locking member **172**. When the operating member **171** rotates to the unlocking position, the shaft locking assembly **170** is in the unlocking state, and the shaft locking member **172** disengages from the shaft locking groove **1411**. When the operating member **171** rotates to the locking position, the shaft locking assembly **170** is in the locking state, and the shaft locking member **172** is embedded in the shaft locking groove **1411**. In some examples, the operating member **171** may be disposed on the second housing **160**. In some examples, the rotation axis of the operating member **171** is collinear with or parallel to the rotation axis of the shaft locking member **172**.

[0148] May possible connection relationships exist between the operating member **171**, the shaft locking member **172**, and the support limiting member **173**. In some examples, the support limiting member **173** is fixedly connected to the second housing **160**, the operating member **171** is fixedly connected to the shaft locking member **172**, and the operating member **171** and the shaft locking member **172** are movably connected to the support limiting member **173** and the second housing **160** and can rotate and translate relative to the support limiting member **173** and the second housing **160**. The following mainly provides an illustrative description of this example. In some other examples, the operating member **171** may be non-fixedly connected to the shaft locking member **172**, and an intermediate piece may exist between the operating member **171** and the shaft locking member **172**, as long as the rotation of the operating member **171** can be converted into the rotational ascent or descent of the shaft locking member **172**. In some other examples, the operating member **171** may be fixedly connected to the support limiting member **173**, and the operating member **171** and the support limiting member **173** are movably connected to the shaft locking member **172** and the second housing **160**, can rotate and translate relative to the shaft locking member **172**, and can rotate relative to the second housing **160**.

[0149] In some examples, the support limiting member **173** has a helical track **1731** along the circumferential direction of the shaft locking member **172**, and the shaft locking member **172** is connected to or formed with a mating portion **1721** that is constrained by the helical track **1731** and is movable along the helical track **1731**. The movement of the mating portion **1721** drives the shaft

locking member **172** to rotate upward or downward. In some examples, the shaft locking member **172** is provided with the mating portion **1721** that protrudes along the radial direction. For example, a mating pin may be inserted into the shaft locking member **172** along the radial direction and used as the mating portion **1721**. The support limiting member **173** may be sleeved on the outside of the shaft locking member **172** and may have an inclined surface along the circumferential direction that can abut against the mating pin. It is to be understood that the helical track is not strictly required to be disposed along a helical line in the circumferential direction of the support limiting member **173**, as long as the helical track can roughly allow the corresponding circumferential and axial movements of the shaft locking member **172** when the shaft locking member **172** rotates and translates. In some examples, the support limiting member **173** is at least partially accommodated in the operating member **171**. Specifically, the operating member **171** is provided with an accommodation groove facing the shaft locking member **172** and the support limiting member **173**, an end of the operating member **171** and/or the shaft locking member **172** may be partially extended into the accommodation groove, and the helical track **1731** may be at least partially accommodated in the accommodation groove.

[0150] In some examples, locking portions **1732** are provided at two ends of the helical track **1731** of the support limiting member **173**. When the mating portion **1721** moves to one of the two ends of the helical track **1731**, the mating portion **1721** is limited by the locking portion **1732** and stopped at the position, thereby maintaining the shaft locking assembly **170** in the locking state or the unlocking state. In some examples, the locking portion **1732** includes a locking groove disposed at any end of the helical track **1731**, and the mating portion **1721** falls into the groove to achieve self-locking. The locking portion **1732** may also include a blocking wall disposed at an end of the helical track **1731** reached by the mating portion **1721** under the drive of the elastic member **174**.

[0151] As shown in FIGS. **37** and **38**, the support limiting member **173** is fixedly mounted on the second housing **160**, and the shaft locking member **172** is in the shape of a rod, penetrates the support limiting member **173**, and then extends into the second housing **160** to act on the shaft locking groove **1411**. The operating member **171** is in the shape of a cap, is sleeved on an end of the support limiting member **173** facing away from the shaft locking grooves **1411**, and is fixedly connected to the shaft locking member **172**. The elastic member **174** sleeved on the shaft locking member **172** is always in the compressed state, an end of the elastic member **174** facing away from the shaft locking grooves **1411** is fixed, and the other end of the elastic member **174** abuts against the shaft locking member **172**. The shaft locking member **172** driven by the elastic member **174** rotates upward or downward since the pin-shaped mating portion **1721** extending radially from the shaft locking member **172** is constrained in the circumferential direction by the helical track **1731**. When the operating member **171** is rotated from the unlocking position to the locking position, the external force for rotating the operating member **171** causes the mating portion **1721** that is on the shaft locking member **172** fixedly connected to the operating member **171**, originally falls into the locking groove **1732** at an end of the helical track **1731**, and is in the self-locking state to be driven out of the locking groove **1732** and disengage from the self-locking state, and under the drive of the elastic member **174** and the constraint of the helical track **1731** on the mating portion **1721** against which the helical track **1731** abuts, the shaft locking member **172** rotates downward and is embedded in the shaft locking groove **1411** below until the mating portion **1721** reaches the other end of the helical track **1731**, is limited by the blocking wall **1732**, and cannot move further. In this manner, the shaft locking assembly **170** is maintained in the locking state.

[0152] When the operating member **171** is rotated from the locking position to the unlocking position, the external force for rotating the operating member **171** causes the shaft locking member **172** fixedly connected to the operating member **171** to overcome the drive of the elastic member **174**, the blocking wall **1732** originally limiting the mating portion **1721** no longer works due to the reverse movement of the mating portion **1721**, and under the constraint of the helical track **1731** on

the mating portion **1721** against which the helical track **1731** abuts, the shaft locking member **172** rotates upward and disengages from the shaft locking groove **1411** until the mating portion **1721** falls into the locking groove **1732** at the other end of the helical track **1731** and cannot move further. In this manner, the shaft locking assembly **170** is maintained in the unlocking state.

[0153] In some examples, a guide surface **1412** is provided between adjacent shaft locking grooves **1411**. In the case where the shaft locking member **172** rotates and translates and the shaft locking groove **1411** is not exactly located above or below the shaft locking member **172**, the shaft locking member **172** may rotate and translate to the guide surface **1412**, slide relative to the shaft locking groove **1411** along the guide surface **1412**, and finally, be embedded in the shaft locking groove **1411**. Specifically, the guide surface **1412** may be an arc surface or an inclined surface between adjacent shaft locking grooves **1411**, and the edge of the guide surface **1412** may be flush with the opening of the shaft locking groove **1411**. The shaft locking disc is used as an example. Multiple shaft locking grooves **1411** may be provided along the radial direction toward the center of the shaft locking disk, adjacent shaft locking grooves **1411** may be spaced at equal distances or angles, and the outer circumferential surface of the shaft locking disk between the shaft locking grooves **1411** may be the guide surface **1412**.

[0154] In addition to being applied to the preceding cut-off saw **100**, the shaft locking assembly **170** including the operating member **171**, the shaft locking member **172**, the support limiting member **173**, and the elastic member **174** may be applied to other types of power tools. The shaft locking grooves are provided on the motor shaft, the output shaft, or the transmission assembly, and the shaft locking assembly **170** can lock and release the output shaft in the manner described above.

[0155] A bearing is disposed between the long shaft **141** and the second housing **160**, and the bearing supports the long shaft **141**. The second housing **160** is provided with a bearing chamber for mounting the bearing. The second housing **160** may be a split structure, thereby facilitating the installation of the transmission assembly **140** and the output shaft **130**. A bearing is disposed between the output shaft **130** and the second housing **160**, and the bearing supports the output shaft **130**.

[0156] Referring to FIGS. **1** to **5** and FIGS. **8** to **20**, the cut-off saw **100** further includes a circuit board assembly **180**. The circuit board assembly **180** includes a circuit board **181** and a heat dissipation member **182**. The circuit board **181** is connected to the heat dissipation member **182**. During use, the circuit board **181** generates heat, the heat is transferred to the heat dissipation member **182**, and the heat dissipation member **182** is used for physically dissipating heat for the circuit board **181**. The housing **110** has an opening, and the heat dissipation member **182** is at least partially exposed relative to the housing **110** through the opening. The heat dissipation member **182** may be exposed to the outside of the housing **110** or aligned with the opening of the housing **110** so that the heat dissipation member **182** can be in contact with the external space for heat exchange, thereby achieving natural heat dissipation and improving the heat dissipation effect.

[0157] The circuit board **181** is disposed on a side of the heat dissipation member **182**, and the circuit board **181** and the heat dissipation member **182** are in surface contact with each other, thereby increasing the contact area and improving the heat dissipation effect. In some examples, the heat dissipation member **182** is a heat dissipation box. In some examples, the heat dissipation member **182** is a heat sink. In some examples, the heat dissipation member **182** is made of aluminum.

[0158] When the heat dissipation member **182** is a heat dissipation box, a side of the heat dissipation box facing away from the circuit board **181** has multiple heat dissipation through holes **1821**, and the heat dissipation through holes **1821** penetrate the box body of the heat dissipation box along a direction parallel to a plane where the circuit board **181** is located, thereby increasing the heat dissipation area of the heat dissipation box and improving the heat dissipation efficiency. In some examples, the circuit board **181** is disposed on the top of the heat dissipation box, the heat dissipation through holes **1821** are disposed on the bottom of the heat dissipation box, and the

bottom surface of the heat dissipation box is exposed to the outside of the housing **110** or aligned with the opening position of the housing **110**. An accommodation cavity for accommodating the circuit board **181** is disposed on the top of the heat dissipation box.

[0159] The circuit board **181** is located below the electric motor **120**, and the projection of the circuit board **181** and the projection of the electric motor **120** in the up and down direction at least partially coincide so that the space is fully utilized and the layout is reasonable.

[0160] The cut-off saw **100** further includes a heat conductive member **190** connected to the housing **110**. The heat conductive member **190** is at least partially located outside the housing **110**. The heat dissipation member **182** is at least partially exposed relative to the housing **110** through the opening and is in thermal contact with the heat conductive member **190**. The heat from the heat dissipation member **182** can be transferred to the heat conductive member **190**, and the heat conductive member **190** is in contact with the external space for heat exchange, thereby achieving natural heat dissipation and further improving the heat dissipation effect.

[0161] In some examples, the opening is located on a front bottom surface of the housing **110**, the heat conductive member **190** is connected to the housing **110** around the opening and has a heat conductive plane covering the opening, and the heat dissipation member **182** is in contact with the heat conductive plane. Through surface-to-surface contact, the contact area between the heat dissipation member **182** and the heat conductive member **190** can be increased, thereby increasing the heat conduction area efficiency and improving the heat dissipation effect.

[0162] The heat conductive member **190** is a wear-resistant metal member to improve the heat dissipation effect. In some examples, the heat conductive member **190** is bent into a U-shape or a V-shape so that the heat conductive member **190** can be in contact with the housing **110**. The edges of the heat conductive member **190** are bent to cover the periphery of the opening of the housing **110**. In some examples, the heat conductive member **190** covers the junction between the bottom and the front of the body portion **111**. The heat conductive member **190** and the housing **110** may be connected via screws. In some examples, the heat conductive member **190** is made of a nickel-iron alloy.

[0163] The heat conductive member **190** not only plays a role in heat conduction but also has a supporting function. The bottom surface of the heat conductive member **190** forms or supports the support surface so that the cut-off saw **100** can be supported when placed horizontally. The heat conductive member **190** is configured to be the wear-resistant metal member, thereby reducing wear and ensuring the heat dissipation and supporting effects.

[0164] The cut-off saw **100** further includes a sealing member **183** disposed around the opening and filling the gaps between the housing **110**, the heat dissipation member **182**, and the heat conductive member **190**. The sealing member **183** is used for preventing water from entering the housing **110**. The sealing member may be a sealing ring, a sealing strip, or another sealing structure. As shown in FIG. **19**, the sealing member **183** is a sealing ring that is made of rubber material, is sleeved on the bottom of the heat dissipation box, and completely fills the gap between the heat dissipation box and the bottom opening of the housing **110**. The sealing member **183** further has positioning and protective edges that extend a certain length along the outer periphery of the heat conductive plane.

[0165] In some examples, the heat conductive member **190**, the heat dissipation member **182**, and the second housing **160** are connected; the heat dissipation member **182**, the heat conductive member **190**, the second housing **160**, and the saw blade **150** are all made of conductive materials, and the static electricity generated in a cutting process is transferred to the saw blade **150** through the heat dissipation member **182**, the heat conductive member **190**, and the second housing **160** and then released to the ground. In some examples, the heat conductive member **190** is fixedly connected to the housing **110**, and the connection between the heat conductive member **190** and the second housing **160** may be abutting or overlapping, thereby facilitating the transfer of static electricity. In some examples, any two of the heat dissipation member **182**, the heat conductive

member **190**, the second housing **160**, and the saw blade **150** may be connected using screws or bolts made of conductive materials.

[0166] In the cut-off saw **100** provided in the present application, not only is heat dissipated through the contact between the heat dissipation member **182** and the heat conductive member **190**, but heat is also dissipated by spraying coolant onto the saw blade **150**, thereby combining natural heat dissipation and water-cooling heat dissipation to improve the heat dissipation effect.

[0167] The cut-off saw **100** further includes a liquid cooling system **210**. The liquid cooling system **210** sprays coolant onto the saw blade **150** when the saw blade **150** is cutting. The coolant moves through the heat conductive member **190** after leaving the saw blade **150**. By spraying coolant onto the heat conductive member **190** to cool the heat conductive member **190**, natural heat dissipation is combined with water-cooling heat dissipation, thereby further improving the heat dissipation effect of the heat conductive member **190** on the circuit board **181** and the like. Normally, the saw blade **150** rotates counterclockwise. When the saw blade **150** rotates, the coolant sprayed by the liquid cooling system **210** is driven by the rotating saw blade **150** and is basically thrown out from the bottom of the saw blade **150** along the tangential direction of the saw blade **150**. The coolant flow is sprayed toward the heat conductive member **190** by inertia. The arrow direction in FIG. **8** shows the spraying direction of part of the coolant.

[0168] The liquid cooling system **210** includes a transfer pipe **211** for transmitting coolant. A groove for the transfer pipe **211** to pass through is formed at the bottom inside the housing **110**, and an extension path of the transfer pipe **211** and the housing **110** are in contact around the periphery of the circuit board assembly **180**. The flow of the coolant in the transfer pipe **211** takes away part of the heat from the circuit board **181**, thereby further improving the water-cooling heat dissipation effect. Part of the transfer pipe **211** can be squeezed into the groove in the housing **110**, and part of the transfer pipe **211** can be limited by a snap catch or clamp so that the position of the transfer pipe **211** is stable.

[0169] The projection of the transfer pipe **211** and the projection of the circuit board **181** in the up and down direction at least partially coincide; and/or the projection of the transfer pipe **211** and the projection of the heat dissipation member **182** in the up and down direction at least partially coincide. While heat is dissipated, the space is fully utilized, the structure is compact, and the layout is reasonable. In some examples, the transfer pipe **211** is disposed above the circuit board **181**. In some examples, the projection of the transfer pipe **211** does not overlap with the projection of the circuit board **181** in the up and down direction, so as to decrease the safety risk to the circuit board assembly **180** for the moisture bringing by the transfer pipe **211**.

[0170] One of the benefits of the present application is described below. The housing has the opening, the heat dissipation member is at least partially exposed relative to the housing through the opening and is in thermal contact with the heat conductive member, the heat dissipation member connected to the circuit board in the cut-off saw may be in contact with the space outside the housing and dissipate heat through the opening on the housing, and the heat dissipation member can be in thermal contact with the heat conductive member to transfer heat to the heat conductive member outside the housing, thereby having an excellent heat dissipation effect.

[0171] The cut-off saw **100** further includes the guard **220**, and the saw blade **150** is partially accommodated in the guard **220**. The liquid cooling system **210** includes a left nozzle **212** and a right nozzle **213**, the left nozzle **212** is disposed on the left end surface of the guard **220**, the right nozzle **213** is disposed on the right end surface of the guard **220**, and the left nozzle **212** and the right nozzle **213** penetrate the guard **220** to output the coolant. Of course, the liquid cooling system **210** of the cut-off saw **100** can use a single nozzle to cool down the saw blade **150**. The examples described below can also be used in cut-off saw with single nozzle.

[0172] In some examples, as shown in FIGS. **17** to **20** and FIG. **30**, the left nozzle **212** may include a first member **2121** and a second member **2122**, and the first member **2121** is connected to the guard **220**. For example, first members **2121** may be connected to the left and right end surfaces of

the guard **220** through snap grooves, a through hole penetrating the guard **220** exists, and the transfer pipe **211** may be connected to the first members **2121** to connect with the through hole and then output the coolant. For example, the first member **2121** has a protrusion on which the transfer pipe **211** is sleeved, and the protrusion is internally penetrated and connected to the through hole. In some examples, the installation of the first member **2121** and the guard **220** features a foolproof design to ensure the correct installation position. The second member **2122** is detachably connected to the first member **2121**. When the second member **2122** is mounted to the first member **2121**, an end of the through hole facing away from the saw blade **150** is blocked and closed by the second member **2122**. When the second member **2122** is disassembled from the first member **2121**, the end of the through hole facing away from the saw blade **150** is opened to the outside. As shown in FIG. **20**, the first member **2121** has a through hole that is collinear with the output shaft **130**, and the through hole has another conduction branch along the circumferential direction for the connection of the transfer pipe **211**. The second member **2122** is a nut threadedly mating with the outer circumferential surface of the first member **2121**. When the nut is tightened, the nut blocks and closes the end of the through hole facing away from the saw blade **150** so that the coolant from the transfer pipe **211** is sprayed unidirectionally onto the saw blade **150**. After the nuts are disassembled, two ends of the through hole are opened, and the user may use needle-shaped or rod-shaped parts to conveniently and effectively clean the dust that is clogged in the through hole. In some examples, the second member **2122** may be of the same model as the nut used when the saw blade **150** is mounted to the output shaft **130** so that the related accessories are more universal and less complicated. In some examples, a waterproof ring is further provided in the second member **2122** or the nut, and the contact surface between the second member **2122** and the first member **2121** and/or the contact surface between the second member **2122** and the guard **220** are sealed and waterproofed by the waterproof ring. Equivalently, in some examples, the right nozzle **213** may also include the first member and the second member. In some examples, the left nozzle **212** and the right nozzle **213** are wear-resistant and easily machinable aluminum parts.

[0173] In some examples, the transfer pipe **211** extends from the front end of the housing **110**, passes through the gap between the second housing **160** and the guard **220**, crosses over the guard **220**, and is connected to the left nozzle **212** and the right nozzle **213**. The transfer pipe **211** and the second housing **160** are fixedly connected, for example, by screws.

[0174] The guard **220** is provided with a guard handle **230** for the user to operate to rotate the guard **220**. When the transfer pipe **211** crosses over the guard **220**, the transfer pipe **211** is limited by the guard handle **230**. The guard handle **230** and the guard **220** may be locked by screws. In some examples, the transfer pipe **211** is sandwiched between the guard **220** and the guard handle **230**. In some examples, the transfer pipe **211** passes through the guard handle **230**.

[0175] The cut-off saw **100** further includes a support guide member **240**. The support guide member **240** has a support portion **241** located between the second housing **160** and the guard **220**. The transfer pipe **211** passes through the support portion **241** and is limited by the support portion **241**. In some examples, the support portion **241** is provided with a groove for the transfer pipe **211** to pass through. In some examples, the support portion **241** is provided with a pipe for the transfer pipe **211** to pass through. The support guide member **240** and the guard **220** may be locked by screws.

[0176] The support guide member **240** further has a guide groove located behind the guard **220**, and part of the guard **220** is inserted into the guide groove, thereby supporting and guiding the guard **220** without affecting the rotation of the guard **220**. In some examples, the support guide member **240** is a split structure to facilitate installation and disassembly. The support guide member **240** includes the support portion **241** and a guide portion **242**. The support portion **241** and the guide portion **242** are disposed on the left and right sides of the guard **220**. The support portion **241** and the guide portion **242** snap together to form the guide groove. The support portion **241** and the guide portion **242** may be threadedly connected to the guard **220**.

[0177] A splash guard **250** is disposed at the bottom of the guard **220**. The liquid cooling system **210** sprays coolant onto the saw blade **150** when the saw blade **150** is cutting. The splash guard **250** can reduce water splashing and prevent water from entering the housing **110**. The splash guard **250** and the guard **220** may be locked by screws. The splash guard **250** and the guide portion **242** may be integrally formed to facilitate machining and production.

[0178] The second housing **160** is located in front of the housing **110**, and the guard **220** is located in front of the housing **110**. The second housing **160** and the guard **220** are arranged along the left and right direction, the support guide member **240** is sandwiched between the second housing **160** and the guard **220**, and the second housing **160**, the support guide member **240**, and the guard **220** are connected to increase the structural strength.

[0179] In some examples, the liquid cooling system **210** further includes a liquid inlet valve **214**. The transfer pipe **211** is used for transferring the coolant, and the liquid inlet valve **214** is disposed on the transfer pipe **211**. The liquid inlet valve **214** is located outside the housing **110**, making it convenient for the user to operate. In some examples, the liquid inlet valve **214** is disposed on the rear side of the grip, and the user may operate the liquid inlet valve **214** when holding the grip. Based on the above, the grip includes the main handle **112** located behind the body portion **111**. In some examples, the liquid inlet valve **214** may be disposed on the lower right side of the main handle **112** and may be directly connected to a water inlet.

[0180] In some examples, the flow rate of the coolant in the transfer pipe **211** is a constant value, and the liquid inlet valve **214** acts as a switch. In some examples, the liquid inlet valve **214** adjusts the flow rate of the coolant in the transfer pipe **211**. The liquid inlet valve **214** and the housing **110** may be fixed by screws. In some examples, the liquid inlet valve **214** is a rotary ball valve, and the valve hole opening of the liquid inlet valve **214** is adjustable, thereby adjusting the flow rate of the coolant in the transfer pipe **211**.

[0181] The cut-off saw **100** provided in the present application further utilizes air cooling for heat dissipation. The cut-off saw **100** further includes a fan **260** mounted on the motor shaft **121**. The housing **110** has an air inlet **114** and an air outlet **115**. The cooling airflow enters the housing **110** from the air inlet **114**, flows through the circuit board assembly **180** and the electric motor **120** in sequence, and is discharged from the air outlet **115** while dissipating the heat from the circuit board assembly **180** and the electric motor **120**.

[0182] When the cooling airflow flows through the circuit board assembly **180**, the cooling airflow passes through the heat dissipation member **182** and takes away heat from the heat dissipation member **182**. In some examples, the heat dissipation member **182** is a heat dissipation box, a side of the heat dissipation box facing away from the circuit board **181** has multiple heat dissipation through holes **1821**, and the heat dissipation through holes **1821** penetrate the box body of the heat dissipation box along the direction parallel to the plane where the circuit board **181** is located. By providing the heat dissipation through holes **1821** on the heat dissipation box, the heat dissipation area can be increased, the air flow rate on the heat dissipation surface can be increased, and the heat dissipation effect can be improved.

[0183] In some examples, the heat dissipation through holes **1821** extend along the left and right direction of the housing **110**. In some examples, the heat dissipation through holes **1821** extend along a straight line. In some examples, the heat dissipation through holes **1821** extend along a curve, thereby extending the heat dissipation path. In some examples, the multiple heat dissipation through holes **1821** are distributed in multiple rows along the front and rear direction or the up and down direction.

[0184] In some examples, the cut-off saw **100** further includes an air guide plate **270** connected to the housing **110** and located inside the housing **110**. The housing **110**, the air guide plate **270**, and the heat dissipation box form a first airflow channel between the air inlet **114** and the heat dissipation through holes **1821**, and the cooling airflow enters the housing **110** from the air inlet **114** and passes through the first airflow channel and the heat dissipation through holes **1821** in

sequence. The first airflow channel is in a basically sealed state between the air inlet **114** and the heat dissipation through holes **1821**, and the air guide plate **270** guides the cooling airflow. By providing the sealed first airflow channel, negative pressure can be formed in the first airflow channel easily, thereby promoting the smooth flow of external air into the housing **110**.

[0185] The shape of the air guide plate **270** may be set according to actual requirements to guide the airflow to the heat dissipation member **182**. In some examples, the air guide plate **270** is stepped. The air guide plate **270** and the housing **110** may be connected by screws.

[0186] The housing **110** has the air outlet **115**, and the electric motor **120** has a motor housing **123**. The housing **110** and the motor housing **123** form a second airflow channel between the electric motor **120** and the air outlet **115**, and the cooling airflow flows through the electric motor **120** in the motor housing **123** and is discharged from the air outlet **115** through the second airflow channel. The second airflow channel is in a basically sealed state between the electric motor **120** or the motor housing **123** and the air outlet **115**. By providing the sealed second airflow channel, negative pressure can be formed in the second airflow channel easily, thereby promoting the smooth flow of external air into the motor housing **123**.

[0187] When the electric motor **120** rotates, the electric motor **120** drives the fan **260** to rotate so that negative pressure can be formed in the first airflow channel and the second airflow channel, and the external airflow enters the housing **110** from the air inlet **114**, passes through the first airflow channel and the heat dissipation through holes **1821** in sequence, enters the motor housing **123**, flows through the electric motor **120** in the motor housing **123**, and is discharged from the air outlet **115** through the second airflow channel. The dashed arrows in FIGS. **11**, **12**, **14**, and **16** show the airflow directions.

[0188] In some examples, the heat dissipation box is located below the electric motor **120**, and the projection of the heat dissipation box and the projection of the electric motor **120** in the up and down direction at least partially coincide so that the cooling airflow passes through the heat dissipation through holes **1821** and then enters the motor housing **123** through the inner cavity of the housing **110**.

[0189] In some examples, the air inlet **114** is disposed at the rear of the housing **110**, and the air outlet **115** is disposed at the front of the housing **110**, thereby guiding the airflow to flow from the rear to the front and extending the flow path of the cooling airflow. The cooling airflow flows through components such as the circuit board **181**, the heat dissipation box, and the electric motor **120** to achieve comprehensive heat dissipation. In some examples, the air inlet **114** is located on a left wall and/or a right wall of a middle and rear part of the housing **110**, and the air outlet **115** is located at the front part of the housing **110**. In some examples, the air outlet **115** is located on the left wall and/or the right wall of the front part of the housing **110**. In some examples, the air outlet **115** is located on the left wall and the right wall of the front part of the housing **110**, and the air outlet **115** is open to two sides, thereby improving the heat dissipation effect.

[0190] In some examples, the air inlet **114** includes multiple air inlet holes spaced apart on the housing **110**, thereby fully utilizing the air inlet space on the housing **110** and increasing the air inlet area. The air inlet hole is circular, elliptical, or rectangular. In some examples, the air outlet **115** includes multiple air outlet holes spaced apart on the housing **110**, thereby fully utilizing the air outlet space on the housing **110** and increasing the air outlet area. The air outlet hole is circular, elliptical, or rectangular.

[0191] The air-cooling heat dissipation path is isolated from the water-cooling heat dissipation path so that the coolant for water cooling does not spray into the interior of the housing **110**.

[0192] Referring to FIGS. **17** to **22**, the cut-off saw **100** further includes a guard adjustment mechanism **280**, and the guard **220** is connected to the second housing **160** through the guard adjustment mechanism **280**. The guard adjustment mechanism **280** includes an adjustment bolt **281** and an adjustment handle **282**. A rotation track **221** is disposed on a side of the guard **220** facing the second housing **160**. An end of the adjustment bolt **281** is fixedly connected to the second

housing **160**, and the other end of the adjustment bolt **281** is rotatably connected to the guard **220** through the rotation track **221**. The adjustment handle **282** releases or locks the relative rotation between the adjustment bolt **281** and the guard **220**, thereby locking or releasing the relative rotation between the second housing **160** and the guard **220**.

[0193] In some examples, the rotation centerline of the guard **220** coincides with the output axis **131** of the output shaft **130**, and the rotation track **221** is in an arc shape around the rotation centerline to facilitate smooth rotation. The adjustment bolt **281** is threadedly connected to the second housing **160**, the adjustment handle **282** is disposed at an end of the adjustment bolt **281**, and the other end of the adjustment bolt **281** can abut against an elastic gasket **283** so that at least part of the elastic gasket **283** is inserted into the rotation track **221**, and the guard **220** is locked using the friction between the elastic gasket **283** and the guard **220**.

[0194] The guard adjustment mechanism **280** performs stepless adjustment on the angle of rotation of the guard **220** about the output shaft **130**. The guard **220** is rotatable about the output shaft **130** within a range of 0° to 45° . FIG. **21** shows the state of the guard **220** at the first position, and FIG. **22** shows the state of the guard **220** at the second position. The guard **220** rotates from the first position to the second position by an angle of 45° .

[0195] Based on the above, the transfer pipe **211** passes through the gap between the second housing **160** and the guard **220**, crosses over the guard **220**, and is connected to the left nozzle **212** and the right nozzle **213**. The guard **220** is adjusted by the guard adjustment mechanism **280** and is rotatable about the output shaft **130** within a certain angle range. The transfer pipe **211** that is connected to one or more of the second housing **160**, the guard **220**, the left nozzle **212**, and the right nozzle **213** may bend during the rotation of the guard **220**, resulting in poor coolant flow. In some examples, as shown in FIG. **29**, an elastic support member **2112** may be provided in at least part of the transfer pipe **211**, thereby supporting the coolant channel space in the pipe and preventing the transfer pipe **211** from bending and blocking the coolant flow when the guard **220** rotates. Specifically, a spring is filled in at least part of the transfer pipe **211** located between the second housing **160** and the guard **220**, the left nozzle **212**, or the right nozzle **213**.

[0196] Referring to FIGS. **1** and **10** and FIGS. **23** to **25**, the cut-off saw **100** further includes a battery pack **290**, and the battery pack **290** supplies power to at least the electric motor **120**. A battery compartment **1111** is formed inside the body portion **111** of the housing **110**, and the battery compartment **1111** is used for mounting the battery pack **290**. To maintain the balance of the whole machine, the electric motor **120** is located on the front side of the body portion **111**, and the battery pack **290** is located on the rear side of the body portion **111**. In the present application, the housing **110** of the cut-off saw **100** includes the body portion **111** and a battery compartment cover **116**. By closing the battery compartment cover **116**, the battery compartment **1111** that is in a sealed state and extends along the left and right direction can be formed in the body portion **111**. The battery pack **290** may be coupled to the battery compartment **1111** along the left and right direction of the housing **110** to adapt to the layout of the whole machine. In some examples, the body portion **111** has a partition separating the front part where the electric motor **120** is located and the battery compartment **1111** where the battery pack **290** is located so that the front part of the body portion **111** that adopts air cooling does not affect the sealing of the battery compartment **1111**.

[0197] To facilitate the replacement of the battery pack **290**, the battery pack **290** and the battery compartment **1111** are detachably connected, specifically, via a snap-fit, a threaded connection, or a magnetic connection. In some examples, one of the battery pack **290** and the battery compartment **1111** is provided with a first snap catch **291**, and the other one of the battery pack **290** and the battery compartment **1111** is provided with a first snap groove. The first snap catch **291** mates with the first snap groove so that the battery pack **290** can be locked to the battery compartment **1111**. In some examples, the first snap catch **291** is disposed on the housing of the battery pack **290**, and the first snap groove is disposed on the inner wall of the battery compartment **1111**.

[0198] The cut-off saw **100** further includes a battery pack ejection mechanism **310** disposed on the

inner wall of the battery compartment **1111**. When the first snap catch **291** disengages from the first snap groove, at least part of the battery pack **290** pops up from the battery compartment **1111** through the battery pack ejection mechanism **310** along the insertion and removal direction.

[0199] The battery pack ejection mechanism **310** includes a ejection member which is elastic. When the battery pack **290** is mounted in the battery compartment **1111**, the ejection member is compressed and deformed. When the first snap catch **291** disengages from the first snap groove, the deformation of the ejection member is restored under the action of the elastic force, thereby applying an external force to the battery pack **290** to cause the battery pack **290** to disengage from the battery compartment **1111**. In this manner, at least part of the battery pack **290** pops up from the battery compartment **1111**, thereby facilitating the removal of the battery pack **290**.

[0200] The battery pack **290** includes a third operating member **294**, and the first snap catch **291** can disengage from the first snap groove by pressing the third operating member **294**. When the battery pack **290** is disassembled, the third operating member **294** may be pressed to disengage the first snap catch **291** from the first snap groove, and the deformation of the ejection member is restored under the action of the elastic force, thereby applying an external force to the battery pack **290** to cause the battery pack **290** to disengage from the battery compartment **1111**. When the battery pack **290** and the battery compartment **1111** are locked, the third operating member **294** is located at a position convenient for operation by the user. In some examples, the first snap catch **219** is disposed on the top of the battery pack **290**, and the third operating member **294** is disposed on the upper left part or upper right part of the battery pack **290** near the battery compartment cover **116**.

[0201] The third operating member **294** moves in unison with the first snap catch **291**, and the first snap catch **291** is elastically connected to the housing of the battery pack **290**. By pressing or releasing the third operating member **294**, the first snap catch **291** is moved so that the first snap catch **291** can disengage from the first snap groove.

[0202] The battery compartment cover **116** has a closed state and an open state. In the closed state, the battery compartment cover **116** seals the battery compartment **1111**. In the open state, the battery compartment **1111** is opened, and the battery pack **290** is detachably inserted into or removed from the battery compartment **1111** along the left and right direction.

[0203] The battery compartment cover **116** and the body portion **111** may be connected via a snap-fit, a threaded connection, or a magnetic connection. In some examples, an end of the battery compartment cover **116** is rotatably connected to the body portion **111**, and the other end of the battery compartment cover **116** is snap-fit with the body portion **111**. In some examples, one of the battery compartment cover **116** and the body portion **111** is provided with a second snap groove **1161**, the other one of the battery compartment cover **116** and the body portion **111** is provided with a second snap catch **1112**, and the second snap catch **1112** mates with the second snap groove **1161**, thereby sealing the battery compartment **1111**. In some examples, the second snap groove **1161** is disposed on the battery compartment cover **116**, and the second snap catch **1112** is disposed on the body portion **111**. In some examples, the battery compartment cover **116** is snap-fit with the body portion **111**, and multiple snap catches are arranged at intervals along the circumferential direction of the battery compartment cover **116**.

[0204] The battery compartment cover **116** is rotatably connected to the body portion **111**. The battery compartment cover **116** is rotatable about a pivot axis, and the pivot axis is perpendicular to the bottom surface of the housing **110**. The bottom of the housing **110** has the support surface for supporting the cut-off saw **100** when the cut-off saw **100** is placed horizontally, and the pivot axis is perpendicular to the support surface. When the battery pack **290** is mounted or removed, the cut-off saw **100** may be placed horizontally so that the support surface supports the cut-off saw **100**, thereby facilitating the installation and removal operations.

[0205] In some examples, the pivot axis extends along the up and down direction of the housing **110**, the battery compartment cover **116** can flip along the front and rear direction, and the battery

compartment cover can be disposed on the left side or the right side of the housing **110**. In some examples, the pivot axis extends along the up and down direction and is set away from the main handle **112**. During the opening process, the battery compartment cover **116** flips from back to front to avoid interfering with the hand of the user holding the main handle **112**. Compared with other settings such as the pivot axis extending along the front and rear direction, the pivot axis extends along the up and down direction so that the user does not need to counteract gravity when opening the battery compartment cover **116**.

[0206] In some examples, the battery compartment cover **116** in the open state can freely rotate about the pivot axis with an angle of 0° to 180° relative to the closed state. In some examples, the battery compartment cover **116** in the open state rotates about the pivot axis to a preset angle relative to the closed state and remains at the preset angle, where the preset angle is greater than or equal to 90° and less than or equal to 180° .

[0207] In some examples, the battery compartment cover **116** is rotatably connected to the body portion **111** via a connecting shaft **117**. The connecting shaft **117** extends along the pivot axis. The connecting shaft **117** is connected to an elastic member **118**. When the second snap catch **1112** disengages from the second snap groove **1161**, the elastic member **118** causes the battery compartment cover **116** to rotate about the pivot axis to the preset angle relative to the closed state and remain at the preset angle. The elastic member **118** may be a torsion spring that can provide a ejection force for opening the battery compartment cover **116** so that the battery compartment cover **116** can be smoothly opened to the preset angle. In some examples, the battery compartment cover **116** may be limited by a limiting protrusion so that the battery compartment cover **116** remains at the preset angle.

[0208] A circuit wire and a terminal block **320** are provided on the inner wall of the battery compartment **1111**, the circuit wire and the terminal block **320** are connected, and the terminal block **320** is used for the plug-in of the terminals of the battery pack **290**. When the battery pack **290** is inserted into the battery compartment **1111**, the terminals of the battery pack **290** are plugged in the terminal block **320**. The cut-off saw **100** further includes a circuit cover plate **330** disposed in the battery compartment **1111**. The circuit cover plate **330** mates with the inner wall of the battery compartment **1111**, thereby limiting the circuit wire and preventing the circuit wire from warping.

[0209] In some examples, the terminal block **320** is disposed on the upper inner wall of the battery compartment **1111**; the circuit wire and the circuit cover plate **330** are disposed on the left inner wall or the right inner wall of the battery compartment **1111** facing the battery compartment cover **116**. In some examples, the terminal block **320** is disposed on the lower inner wall, the left inner wall, or the right inner wall of the battery compartment **1111**.

[0210] In some embodiments, a bus current of the cutting saw **100** is greater than or equal to 70 A, that is, the power supply current of the battery pack **290** is greater than or equal to 70 A, and the cable size of the bus used by the battery pack **290** in the cutting saw **100** to supply power to the motor **120** and the circuit board assembly **180** and other components through the terminal block **320** is greater than or equal to 12 AWG (American Wire Gauge). It can be understood that under the above-mentioned American Wire Gauge standard, the larger the number, the smaller the wire diameter. The bus cable size greater than or equal to 12 AWG described above includes cable sizes of 10 AWG, 8 AWG, etc. Equivalently, the cable diameter of the bus used by the battery pack **290** in the cutting saw **100** to supply power to the motor **120** and the circuit board assembly **180** and other components (single, solid, round wire diameter) is greater than or equal to 2.053 mm (or 0.0808 inch). In some embodiments, the bus current of the cutting saw **100** is greater than or equal to 80 A, and the bus cable size in the cutting saw **100** is greater than or equal to 10 AWG, that is, the bus cable diameter is greater than or equal to 2.588 mm (or 0.1019 inch); thereby adapting to the high-power working conditions of the cutting saw **100** to provide better current conduction capability and heat dissipation effect.

[0211] In some examples, the weight of the cut-off saw **100** is between 6.0 kg and 7.5 kg when the

battery pack **290** is included, and the weight of the cut-off saw **100** is between 4.5 kg and 5.5 kg when the battery pack **290** is not included.

[0212] In some examples, the weight of the battery pack **290** is between 1.0 kg and 2.0 kg. When the output voltage of the battery pack **290** is 18 V, the output current is between 60 A and 155 A. When the output voltage of the battery pack **290** is 24 V, the output current is between 55 A and 130 A.

[0213] In some examples, the dimension of the battery compartment **111** is between 90 mm and 130 mm along the up and down direction of the housing **110**, between 90 mm and 130 mm along the front and rear direction of the housing **110**, and between 140 mm and 170 mm along the left and right direction of the housing **110**. The preceding dimension of the battery compartment **111** is the inner dimension.

[0214] In some examples, the output power of the electric motor **120** is between 1.2 kW and 2.4 kW. In some examples, the output power of the electric motor **120** is between 1.4 kW and 2.0 kW.

[0215] The cut-off saw **100** further includes the liquid cooling system **210**. The liquid cooling system **210** sprays coolant onto the saw blade **150** when the saw blade **150** is cutting. The liquid cooling system **210** includes the transfer pipe **211** for transferring coolant. In some examples, the projection of the transfer pipe **211** and the projection of the battery compartment cover **116** in the left and right direction do not coincide, thereby preventing the arrangement of the transfer pipe **211** from affecting the opening or closing of the battery compartment cover **116**.

[0216] In some examples, the water inlet of the transfer pipe **211** is disposed on the rear side of the grip, and the transfer pipe **211** extends from the rear to the front at the lower part of the housing **110** and avoids the battery compartment cover **116** to prevent interference with the opening or closing of the battery compartment cover **116**.

[0217] The housing of the battery pack **290** is provided with an indicator **292** for displaying the state information of the battery pack **290**. The housing **110** has a transparent portion **119**, and the transparent portion **119** is aligned with the indicator **292** so that the state information displayed by the indicator **292** can be observed through the transparent portion **119**. The operator can clearly observe the state information.

[0218] In some examples, a window is provided on the housing **110**, and a transparent cover is disposed at the window, thereby forming the transparent portion **119**. In some examples, the transparent portion **119** is detachably connected to the housing **110** for easy replacement and cleaning. The transparent portion **119** and the housing **110** may be connected via a snap-fit, a screw connection, or a magnetic connection.

[0219] In some examples, the indicator **292** displays at least the remaining power information of the battery pack **290**. In some examples, the indicator **292** is the remaining power indicator light for the battery pack **290**. In some examples, the indicator **292** is a display screen. The indicator **292** may display numbers to accurately show the remaining power percentage of the battery pack **290**. Alternatively, the indicator **292** may display the remaining power range through color changes. In some examples, green indicates sufficient power, orange indicates low power, and red indicates very low power.

[0220] The housing **110** includes the body portion **111**, the grip, and the battery compartment cover **116**. The grip is at least partially located behind the body portion **111**. The battery compartment cover **116** seals the battery compartment **111**. The transparent portion **119** is located on the body portion **111**; and/or the transparent portion **119** is located on the battery compartment cover **116**. In some examples, the transparent portion **119** is located on the upper rear side of the body portion **111** and in front of the grip. Even if the user holds the grip, the transparent portion **119** is not blocked, which is convenient for the user to observe.

[0221] The housing **110** is provided with a first operating member **340**, and the first operating member **340** is operated by the user to make the indicator **292** display the state information. When the first operating member **340** is triggered, the indicator **292** displays the state information. When

the first operating member **340** is not triggered, the indicator **292** does not display the state information, thereby saving power. The first operating member **340** is operated by the user, for example, pressed, toggled, or rotated by the user.

[0222] In some examples, the first operating member **340** is triggered, and the indicator **292** displays the state information. After the state information is displayed for a first set duration, the indicator **292** does not display the state information. When the indicator **292** is the remaining power indicator light, the first operating member **340** is triggered, the remaining power indicator light is turned on, and after the remaining power indicator light is turned on for the first set duration, the remaining power indicator light is turned off.

[0223] Further, the housing of the battery pack **290** is provided with a second operating member **293** for the user to operate so that the indicator **292** displays the state information. The second operating member **293** is connected to the first operating member **340**. When the user operates the first operating member **340** externally, the first operating member **340** acts on the second operating member **293** to make the indicator **292** display the state information, thereby partially reusing the original information display function of the battery pack **290**.

[0224] In some examples, the first operating member **340** is disposed on the battery compartment cover **116**. When the battery compartment cover **116** is in the closed state, the first operating member **340** is connected to the second operating member **293** through a rod-shaped part so that when the user presses the first operating member **340**, the second operating member **293** is pressed, thereby displaying the power level. In some examples, the first operating member **340** is elastically connected to the housing **110**, thereby facilitating the pressing operation on the first operating member **340**.

[0225] In some examples, referring to FIGS. **26** to **28**, a battery indicator **1191** may be directly provided on the housing **110** to display the state information of the battery pack **290**, without the need to observe the battery pack **290** through a transparent window. The battery indicator **1191** may display at least the remaining power information of the battery pack **290** and may further display various fault warning information of the battery pack **290**. The battery indicator **1191** may use components including, but not limited to, the remaining power indicator light and the display screen, to achieve information display. In some examples, the battery indicator **1191** further has an operating portion **1192** such as a button, a key, or a touchscreen. In response to the user pressing or clicking the operating portion **1192** or other operations, the battery indicator **1191** can display the state information such as the remaining power. In some examples, the battery indicator **1191** may be disposed at the rear end of the body portion **111** of the housing **110** and on the front side of the grip to facilitate user observation. Of course, the battery indicator **1191** may be disposed at other positions.

[0226] As shown in FIG. **28**, the battery indicator **1191** includes the operating portion **1192** and an indicator portion **1194** that are disposed on the housing **110** and further includes a sub-circuit board **1193** disposed in the housing **110**. The sub-circuit board **1193** may be electrically connected to the operating portion **1192** and the indicator portion **1194** of the battery indicator **1191** and the circuit board **181** of the cut-off saw **100** described above. When the operating portion **1192** is operated by the user, an activation signal is transmitted to the sub-circuit board **1193**. In response to the signal, the sub-circuit board **1193** acquires the state information of the battery pack **290** such as the remaining power information from the circuit board **181** and correspondingly triggers the indicator portion **1194** to display the corresponding information. For example, depending on the remaining power, the corresponding number of indicator lights are turned on or the corresponding power value is displayed, and a red light is turned on or flashes to warn of over-temperature, under-current, low power, or the like. The operating portion **1192** and the indicator portion **1194** may be embedded in the upper rear end of the body portion **111** of the housing **110** and on the front side of the main handle **112**. The sub-circuit board **1193** may be disposed on the back side of the operating portion **1192** and the indicator portion **1194**. The sub-circuit board **1193** and the circuit board **181**

may be electrically connected by a flat cable, and the flat cable may be attached to and fixed on the inner wall of the housing **110**.

[0227] Referring to FIGS. **33** and **34**, to facilitate storage or rest during work breaks for the user, as well as for hanging and draining, the cut-off saw **100** further includes a hook assembly **410** through which the cut-off saw **100** can be stably suspended in a basically vertical posture at one or more support points, including, but not limited to, being suspended on a wall, a box, or a cabinet. In some examples, as shown in FIG. **33**, the hook assembly **410** may be disposed on the bottom surface or side surface of the body portion **111** of the housing **110** of the cut-off saw **100**, for example, on the bottom surface of the battery compartment **1111**. In some other examples, the hook assembly **410** may be disposed near the main handle **112** of the housing **110**, including being disposed at the rear end of the main handle **112** or on the bottom surface of the main handle **112**. Of course, the possibility of disposing the hook assembly **410** at other positions on the housing **110** of the cut-off saw **100** is not excluded. In some examples, the hook assembly **410** is disposed at a higher portion of the bottom surface of the housing **110** along the up and down direction. In some other examples, an accommodation groove may be provided inward at the position on the housing **110** where the hook assembly **410** is mounted, and a hook body **411** and a mounting member **412** are disposed in the accommodation groove, thereby avoiding the influence of the hook assembly **410** on the stability of the cut-off saw **100** placed horizontally.

[0228] In some examples, the hook assembly **410** may have a storage state and a suspended state. The hook assembly **410** may include the hook body **411** and the mounting member **412**. The hook body **411** may be mounted on the housing **110** of the cut-off saw **100** through the mounting member **412**. The mounting member **412** may be fixedly connected to the housing **110**, and the hook body **411** may be rotatably connected to the mounting member **412**. In the storage state and the suspended state, the hook body **411** may be connected to the mounting member **412** at different angles so that the cut-off saw **100** can be in different postures relative to the hook body **411**.

[0229] In some examples, the hook body **411** in the suspended state may be rotated by at least 90° relative to the hook body **411** in the storage state. FIGS. **33** and **34** are used as examples. The hook body **411** and the mounting member **412** are attached to the bottom surface of the battery compartment **1111** which is higher than the bottom surface of the housing **100**. A rotation axis about which the hook body **411** rotates relative to the mounting member **412** is basically parallel to the bottom surface of the battery compartment **1111**. For example, the rotation axis extends along the front and rear direction or the left and right direction. The plane where the hook body **411** is located and the suspended body extends is basically parallel to the bottom surface of the battery compartment **1111** during storage and may be rotated by about 90° during suspension. The surface where the hook body **411** in the suspended state is located is basically perpendicular to the bottom surface of the battery compartment **1111**. The surface where the hook body **411** is located is basically parallel to the ground during storage/suspension. Alternatively, the hook assembly **410** may be disposed at the rear end or bottom of the main handle **112**, the hook body **411** may be rotated by about 90° or 180° during suspension. When the hook body **411** is rotated by 180°, the surface where the hook body **411** is located can be substantially parallel to the bottom surface of the housing **110** of the cutting saw **100** when stored/suspended. In some examples, similar to the battery compartment cover **116** described above, the hook assembly **410** may also include components such as a torsion spring so that the hook body **411** can remain at a preset angle after being rotated.

[0230] It is to be understood that the technical solutions described in the preceding examples can be freely combined without contradiction.

[0231] The basic principles, main features, and advantages of this application are shown and described above. It is to be understood by those skilled in the art that the aforementioned examples do not limit the present application in any form, and all technical solutions obtained through

equivalent substitutions or equivalent transformations fall within the scope of the present application.

Claims

1. A cut-off saw, comprising: a housing having an opening; an electric motor having a motor shaft; an output shaft for mounting and driving a saw blade; a transmission assembly for transmitting power between the motor shaft and the output shaft; a heat conductive member connected to the housing and at least partially located outside the housing; and a circuit board assembly comprising a circuit board connected to a heat dissipation member, wherein the heat dissipation member is at least partially exposed relative to the housing through the opening and is in thermal contact with the heat conductive member.
2. The cut-off saw of claim 1, wherein the opening is located on a front bottom surface of the housing, the heat conductive member is connected to the housing around the opening and has a heat conductive plane covering the opening, and the heat dissipation member is in contact with the heat conductive member through the heat conductive plane.
3. The cut-off saw of claim 1, further comprising a liquid cooling system, wherein the liquid cooling system sprays coolant onto the saw blade when the cut-off saw is cutting, and the coolant moves through the heat conductive member after leaving the saw blade.
4. The cut-off saw of claim 3, wherein the liquid cooling system comprises a liquid inlet valve and a transfer pipe, wherein the transfer pipe is configured to transfer the coolant, and the liquid inlet valve is configured to adjust a flow rate of the coolant in the transfer pipe.
5. The cut-off saw of claim 1, further comprising a sealing member, wherein the sealing member is disposed around the opening and fills gaps between the housing, the heat dissipation member, and the heat conductive member.
6. The cut-off saw of claim 1, wherein the heat conductive member is a wear-resistant metal member.
7. The cut-off saw of claim 1, wherein the heat dissipation member is a heat dissipation box, a side of the heat dissipation box facing away from the circuit board has a plurality of heat dissipation through holes, and the plurality of heat dissipation through holes penetrate a box body of the heat dissipation box along a direction basically parallel to a plane where the circuit board is located.
8. The cut-off saw of claim 7, further comprising a fan, wherein the fan is mounted on the motor shaft, the housing has an air inlet and an air outlet, and a cooling airflow enters the housing from the air inlet, flows through the circuit board assembly and the electric motor in sequence, and is discharged from the air outlet.
9. The cut-off saw of claim 8, further comprising an air guide plate connected to the housing and located inside the housing, wherein the housing, the air guide plate, and the heat dissipation box form a first airflow channel between the air inlet and the plurality of heat dissipation through holes, and the cooling airflow enters the housing from the air inlet and passes through the first airflow channel and the plurality of heat dissipation through holes in sequence.
10. The cut-off saw of claim 8, wherein the electric motor has a motor housing, the housing and the motor housing form a second airflow channel between the electric motor and the air outlet, and the cooling airflow flows through the electric motor in the motor housing and is discharged from the air outlet through the second airflow channel.
11. The cut-off saw of claim 9, wherein the first airflow channel is basically sealed between the air inlet and the plurality of heat dissipation through holes, and/or the second airflow channel is basically sealed between the electric motor and the air outlet.
12. The cut-off saw of claim 8, wherein the air inlet is located on a left wall and/or a right wall of a middle and rear part of the housing, and the air outlet is located at a front part of the housing.
13. The cut-off saw of claim 7, wherein the heat dissipation box is located below the electric motor,

a projection of the heat dissipation box and a projection of the electric motor in an up and down direction at least partially coincide, and a cooling airflow passes through the plurality of heat dissipation through holes and then enters a motor housing through an inner cavity of the housing.

14. The cut-off saw of claim 1, wherein the circuit board is located below the electric motor, and a projection of the circuit board and a projection of the electric motor in an up and down direction at least partially coincide.

15. The cut-off saw of claim 1, further comprising a battery pack, wherein the battery pack supplies power to at least the electric motor, the housing comprises a body portion and a battery compartment cover, a battery compartment is formed inside the body portion, the battery compartment cover has a closed state and an open state, in the closed state, the battery compartment cover seals the battery compartment, and, in the open state, the battery pack is detachably inserted into or removed from the battery compartment along a left and right direction.

16. A cut-off saw, comprising: a housing having an opening; an electric motor having a motor shaft; an output shaft for mounting and driving a saw blade; a transmission assembly for transmitting power between the motor shaft and the output shaft; and a circuit board assembly comprising a circuit board connected to a heat dissipation member, wherein the heat dissipation member is at least partially exposed relative to the housing through the opening.

17. The cut-off saw of claim 16, wherein the housing comprises a body portion extending along a front and rear direction, the body portion has a support surface for supporting the cut-off saw when the cut-off saw is placed horizontally, the motor shaft extends along a motor axis, and the motor axis obliquely intersects with the support surface.

18. The cut-off saw of claim 16, further comprising a second housing, wherein the second housing is connected to the housing and is located outside the housing, the transmission assembly is at least partially located in the second housing, and a central axis of the second housing obliquely intersects with a support surface.

19. The cut-off saw of claim 18, further comprising a heat conductive member, wherein the heat conductive member, the heat dissipation member, and the second housing are connected, the heat dissipation member, the heat conductive member, the second housing, and the saw blade are all made of conductive materials, and static electricity generated in a cutting process is transferred to the saw blade through the heat dissipation member, the heat conductive member, and the second housing and then released to the ground.

20. A cut-off saw, comprising: a housing having an air inlet; an electric motor having a motor shaft; an output shaft for mounting and driving a saw blade; and a transmission assembly for transmitting power between the motor shaft and the output shaft; a circuit board assembly comprising a circuit board, a heat dissipation box, and a plurality of heat dissipation through holes, wherein the circuit board is connected to the heat dissipation box, and the plurality of heat dissipation through holes penetrate a box body of the heat dissipation box along a direction basically parallel to a plane where the circuit board is located; and an air guide plate connected to the housing and located inside the housing, wherein the housing, the air guide plate, and the heat dissipation box form a first airflow channel between the air inlet and the plurality of heat dissipation through holes, and a cooling airflow enters the housing from the air inlet and passes through the first airflow channel and the plurality of heat dissipation through holes in sequence.
